

Collaborators:

Determine the Undetermined for Linear Systems

- If the non-homogeneous term is a constant vector, then what is an appropriate trial solution for $\begin{bmatrix} x_p \\ y_p \end{bmatrix}$?
- What is the important step before you solve for the particular solution after you write an initial trial solution?
- Solve for the particular solution by following the standard process for the method of undetermined coefficients. Since this is a system, you may need to use the row reduction algorithm to solve for the coefficients.

2. A Non-Constant Vector for the Non-Homogeneous Term

Consider this *non-homogeneous linear system of 1st-order ODEs with constant coefficients*:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t \\ -t \end{bmatrix}.$$

Note the non-homogeneous term $\begin{bmatrix} t \\ -t \end{bmatrix}$, a vector function of t , added to the system. Because of the superposition principle, the full solution is of the form $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_h \\ y_h \end{bmatrix} + \begin{bmatrix} x_p \\ y_p \end{bmatrix}$ where $\begin{bmatrix} x_h \\ y_h \end{bmatrix}$ is the homogeneous solution and $\begin{bmatrix} x_p \\ y_p \end{bmatrix}$ is the particular solution.

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| <p>a. Before you solve for the particular solution, solve for the homogeneous solution $\begin{bmatrix} x_h \\ y_h \end{bmatrix}$ using the method of eigenvalues and eigenvectors.</p> | <p>c. Solve for the particular solution by following the standard process for the method of undetermined coefficients. Since this is a system, you may need to use the row reduction algorithm to solve for the coefficients.</p> |
| <p>b. Based on the table of trial solutions for the method of undetermined coefficients, what is an appropriate trial solution for $\begin{bmatrix} x_p \\ y_p \end{bmatrix}$? Then, decide whether multiplying it by t is appropriate.</p> | <p>e. Write the general solution.</p> |